

Part 1: Market Mechanics & Order Books

Understanding how prices are formed in electronic markets.

1. Types of Orders

- **Market Order (Aggressive):** Execute *immediately* at the best available price.
 - Buy Market Order → Pay the **Ask** price.
 - Sell Market Order → Receive the **Bid** price.
- **Limit Order (Passive):** Execute only at a specific price (or better).
 - These orders sit in the **Limit Order Book (LOB)** waiting to be filled.

2. The Bid-Ask Spread

Key Formula

$$\text{Spread} = \text{Best Ask Price} - \text{Best Bid Price}$$

The spread represents the transaction cost for immediate execution.

Exercise: Order Book Dynamics

Consider the following Limit Order Book for **NVIDIA (NVDA)**:

Bid (Buy Orders)		Ask (Sell Orders)	
Volume	Price	Price	Volume
100	\$100.00	\$100.05	500
200	\$99.95	\$100.10	200
50	\$99.90	\$100.15	100

(a) What is the current Bid-Ask Spread?

(b) A trader submits a **Market Buy order for 600 shares**.

- How many shares are bought at what price?

(c) After the trade in (b), what is the **new** Best Ask Price?

Part 2: Forward Contracts

Introduction to linear derivatives.

Definitions

A **Forward Contract** is an agreement to buy (Long) or sell (Short) an asset at time T for price K .

Payoff Formulas at Maturity (T)

$$\text{Payoff for Long} = S_T - K$$

$$\text{Payoff for Short} = K - S_T$$

Example Problem (Preparation for Homework 1, Problem 1)

You enter into a **Short Forward Contract** to sell 100 ounces of Gold at a delivery price of $K = \$2,000$ per ounce. The contract matures in 3 months.

(a) **Scenario 1:** In 3 months, the spot price of Gold is $S_T = \$1,900$.

- Calculate your total payoff.
- Did you make money or lose money? Explain intuitively why.

(b) **Scenario 2:** In 3 months, the spot price of Gold is $S_T = \$2,150$.

- Calculate your total payoff.